

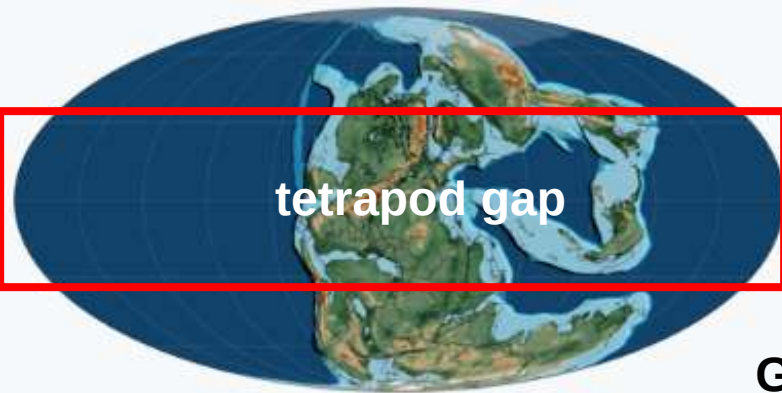
EES 102: Introduction to Environmental Sciences

Instructors – Prof. Baerbel Sinha and Dr. Sunil A. Patil



Image: <https://wiki.seg.org/index.php?curid=13659>

PERMIAN-TRIASSIC EXTINCTION EVENT



- 400 ppmv to 2500 ppmv over 2 Million years PT-Extinction 0.1 ppmv per century
- Antropocene 280 ppmv (1750) to 430 ppmv (2025) 0.9 ppmv/century.
- Rise in the last decade: 3.6 ppmv/year

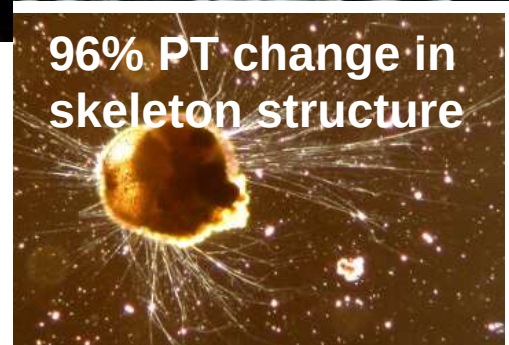
Global temperatures shot up from 18°C to 35°C



96% PT vs 83.9% affected in May 2025 bleaching



96% PT change in skeleton structure



2 Million years

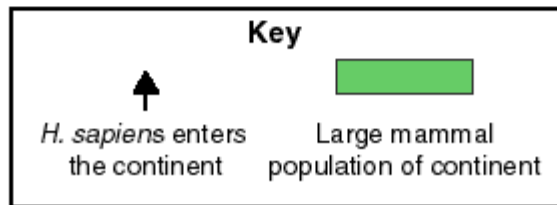
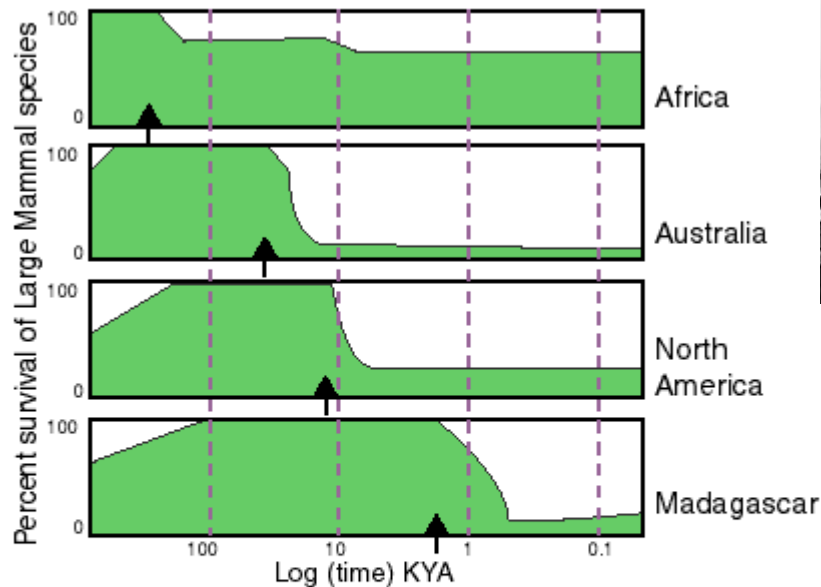
A NEW GEOLOGICAL SUBDIVISION SHOULD BE BASED ON HAVING A GEOCHEMICAL IMPRINT IN AS MANY SYSTEMS AS POSSIBLE

**WHEN GEOLOGISTS DRAW UP A STRATIGRAPHY
THEY NEED TO SEE THE BOUNDARY BETWEEN ONE
AND THE OTHER ERA/PERIOD/EPOCH IN THE ROCK
RECORD**

**IT IS ALWAYS GOOD IF THE TRACE IS SOMETHING
THAT SHOWS UP IN ALL TYPES OF ROCKS. A
GLOBAL IRIIDIUM LAYER CAN BE A GREAT THING.**

MILESTONES OF MASS EXTINCTION

Megafauna extinction



Industrial revolution



Ocean acidification shows in marine sediments particularly through proxy with carbonaceous exoskeleton

Nitrogen cycle perturbation show up in terrestrial ecosystems

The great Acceleration

Post WW2

Nuclear tests



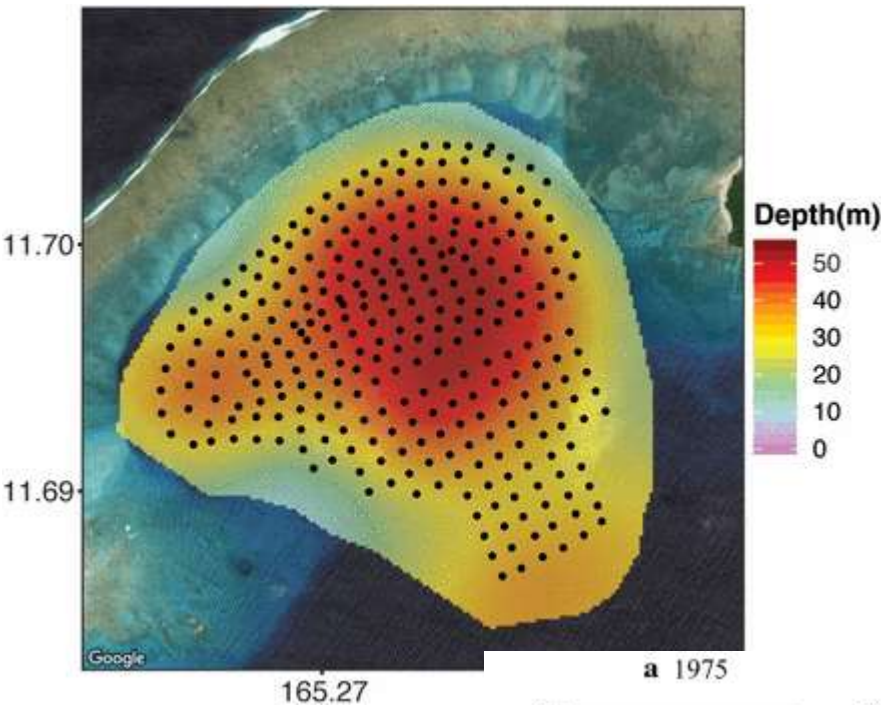
Some terrestrial sediments

Radiation is the best marker

NUCLEAR TEST

Lake sediments from England.

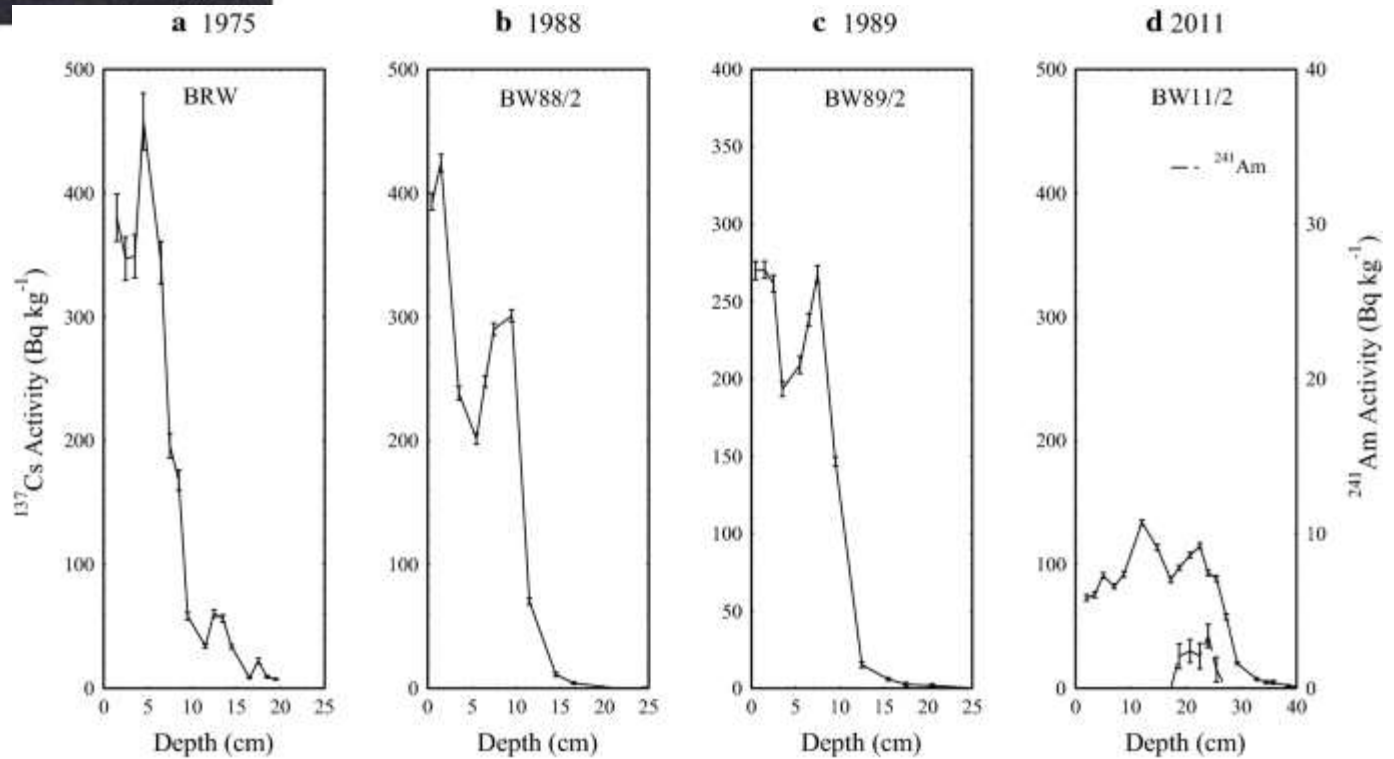
Nuclear tests peaked in 1963 and in 1986 there was the Chernobyl accident



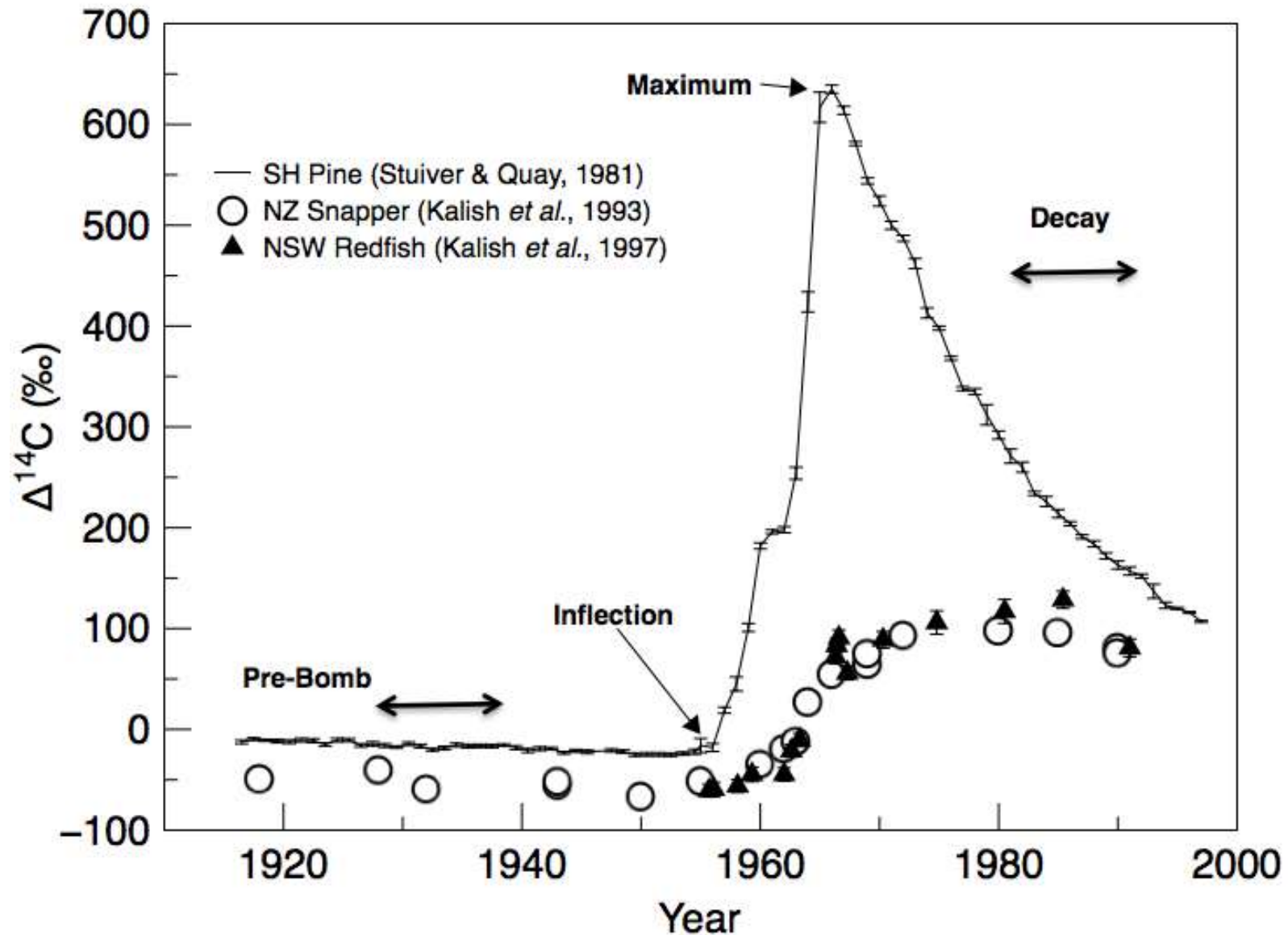
^{137}Cs half life is 70 days

^{241}Am half life is 432.2 years

^{14}C is 5,730 years



THE ^{14}C BOMB PEAK



RADIO NUCLIDES SHOW UP IN MANY SYSTEMS

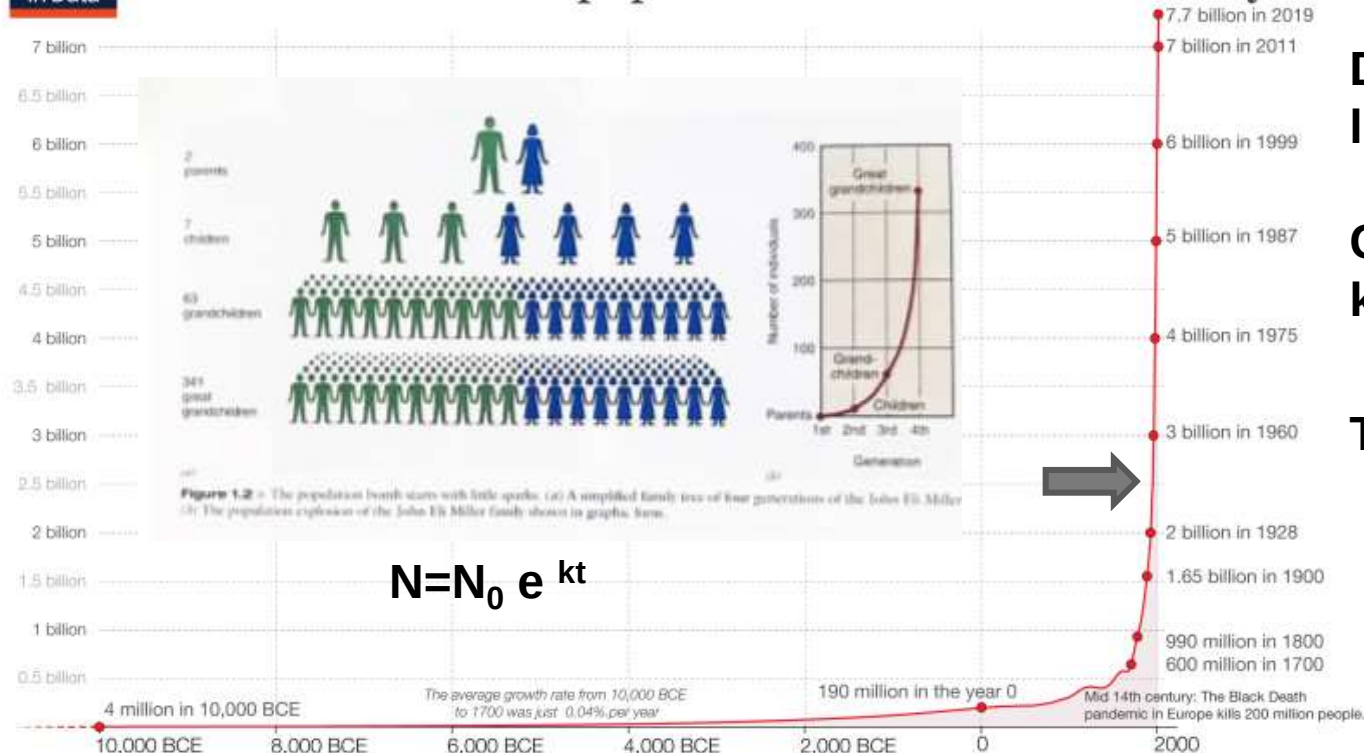
6.3 billion in 2003
growing at 1.36% pa by
2020 yields

$$N = (6.3 \times 10^9 \times e^{(0.0136 \times 17)}) = 7.94 \times 10^9$$

- And the bomb tests coincides with the great acceleration post WW2

Our World
in Data

The size of the world population over the last 12,000 years



$$N = N_0 e^{kt}$$

Doubling time:
 $\ln 2 = kT_d$

Growth rate R (%)
 $k = R/100$

$$T_d = 69.3/R$$

Based on estimates by the History Database of the Global Environment (HYDE) and the United Nations. On OurWorldinData.org you can download the annual data.

This is a visualization from OurWorldinData.org, where you find data and research on how the world is changing.

Licensed under CC-BY-SA by the author Max Roser.

- **E.O. Wilson “Before humans existed, the species extinction rate was (very roughly) one species per million species per year. Estimates for current species extinction rates range from 100 to 10,000 times that, but most hover close to 1,000 times prehuman levels (0.1% per year)**
- **Palumbi (Science, 7 September 2001): mankind also effects evolutionary change in other species, especially in, commercially important, pest and disease organisms, through antibiotics and pesticides.**

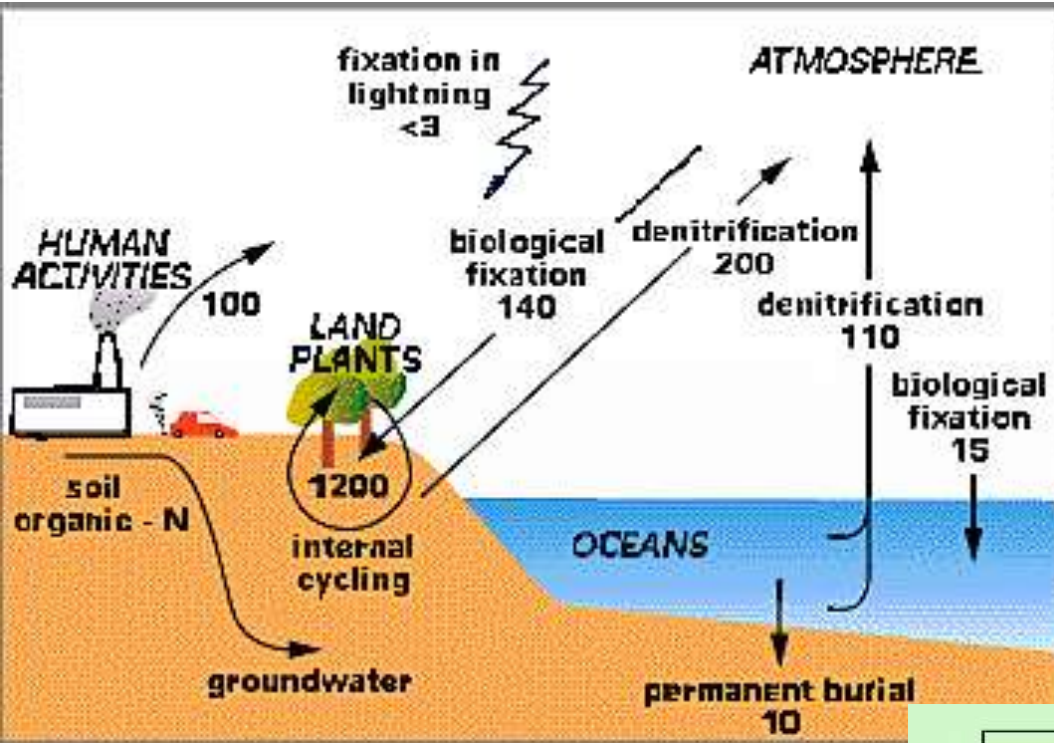
SOURCE: PROF. CRUTZEN

Man the Eroder

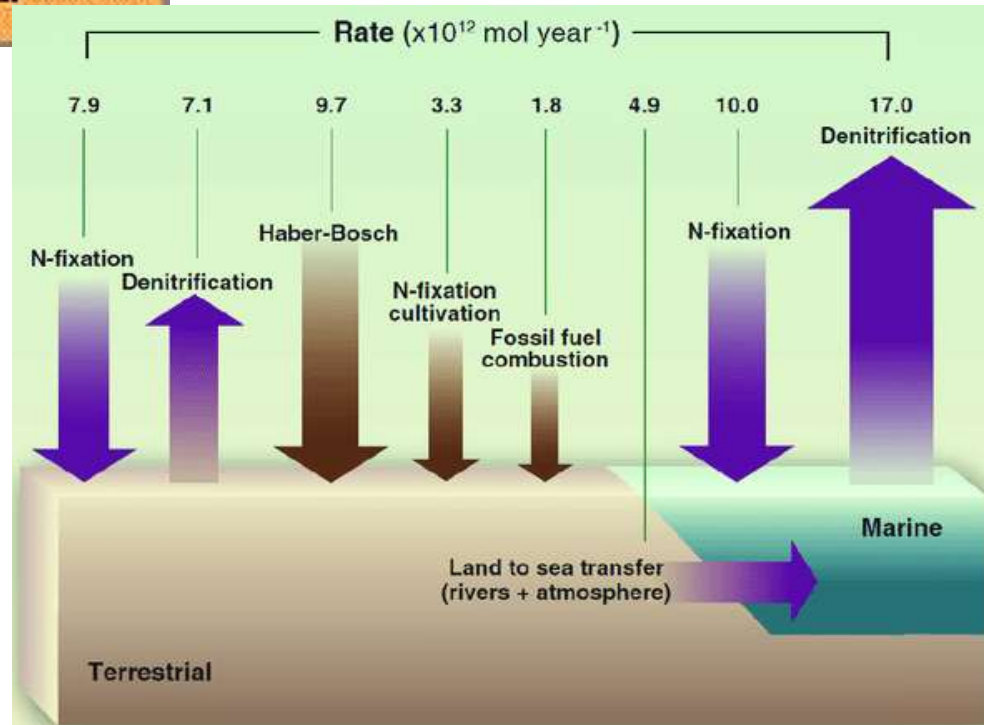
- **Sedimentary rock formation over 500 million years corresponds to an erosion rate of 24 meters per million years.**
- **Man-caused erosion (crop tillage, land conversion for grazing and construction): 15 times natural erosion**
- **At current rate anthropogenic soil erosion would fill the Grand Canyon in 50 years.**

According to Wilkinson (Geology) March 2005.

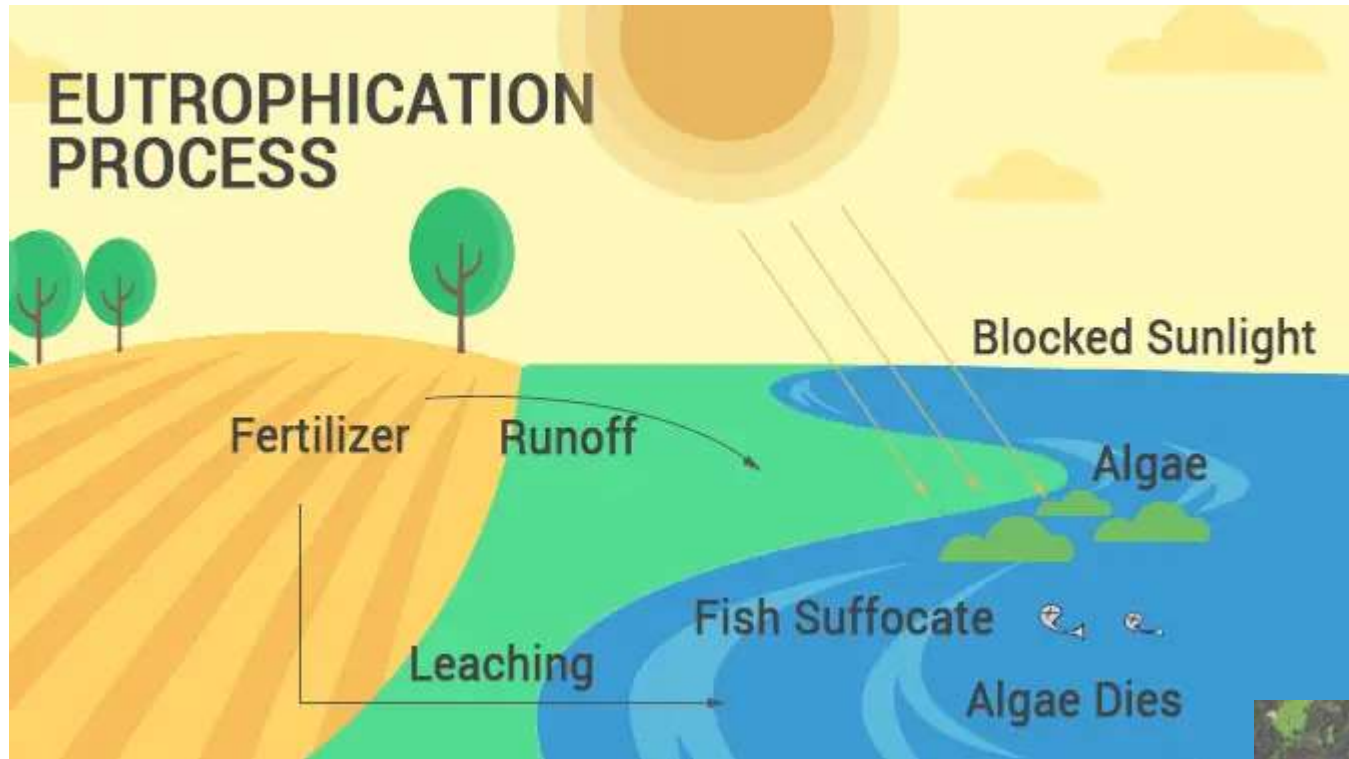
NITROGEN CYCLE



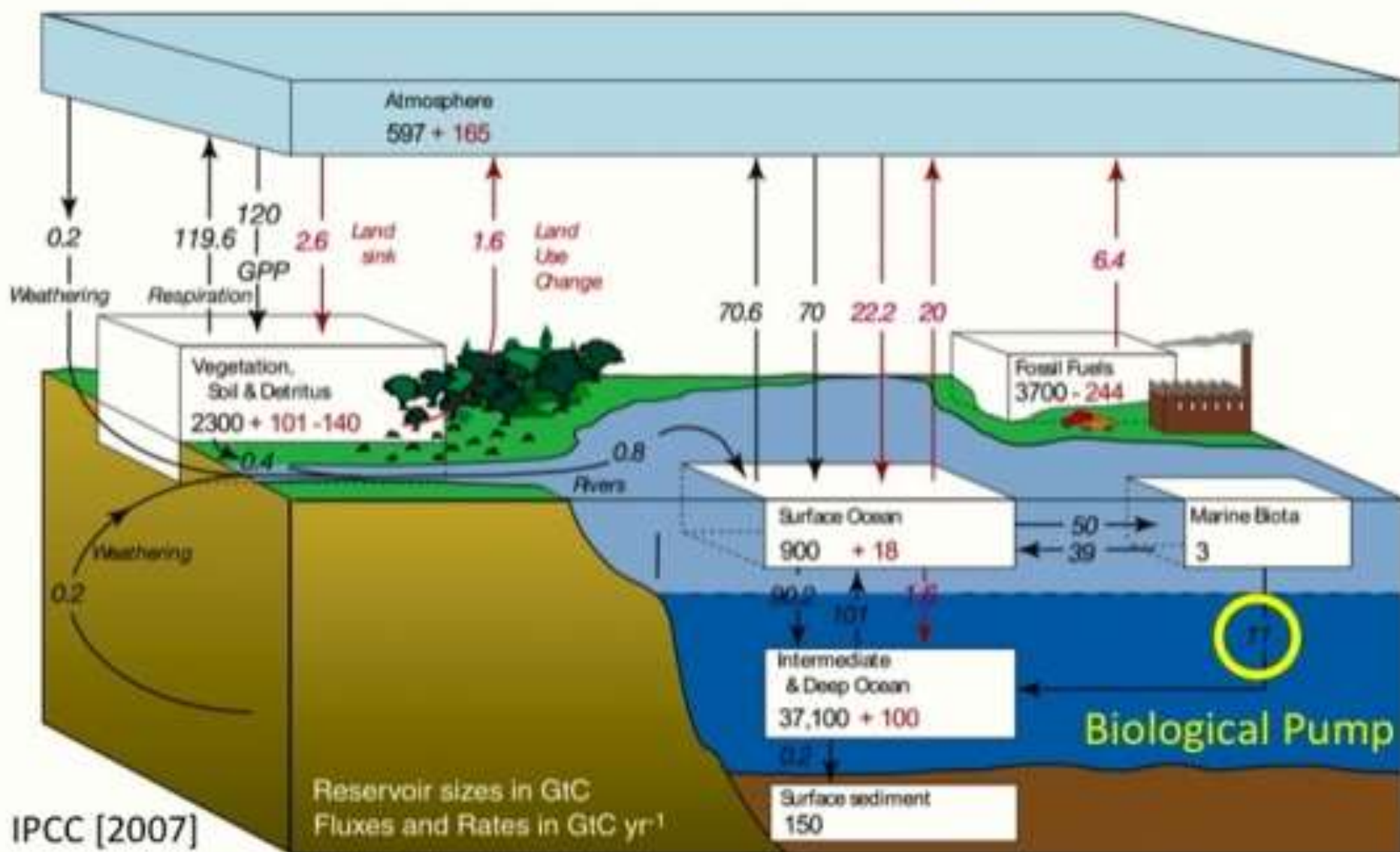
The nitrogen fixation by humans is twice as large as the natural N-fixation
 Haber-Bosch fixes 3x more nitrogen than cultivation could ever fix!



EUTROPHICATION



Global Carbon Cycle



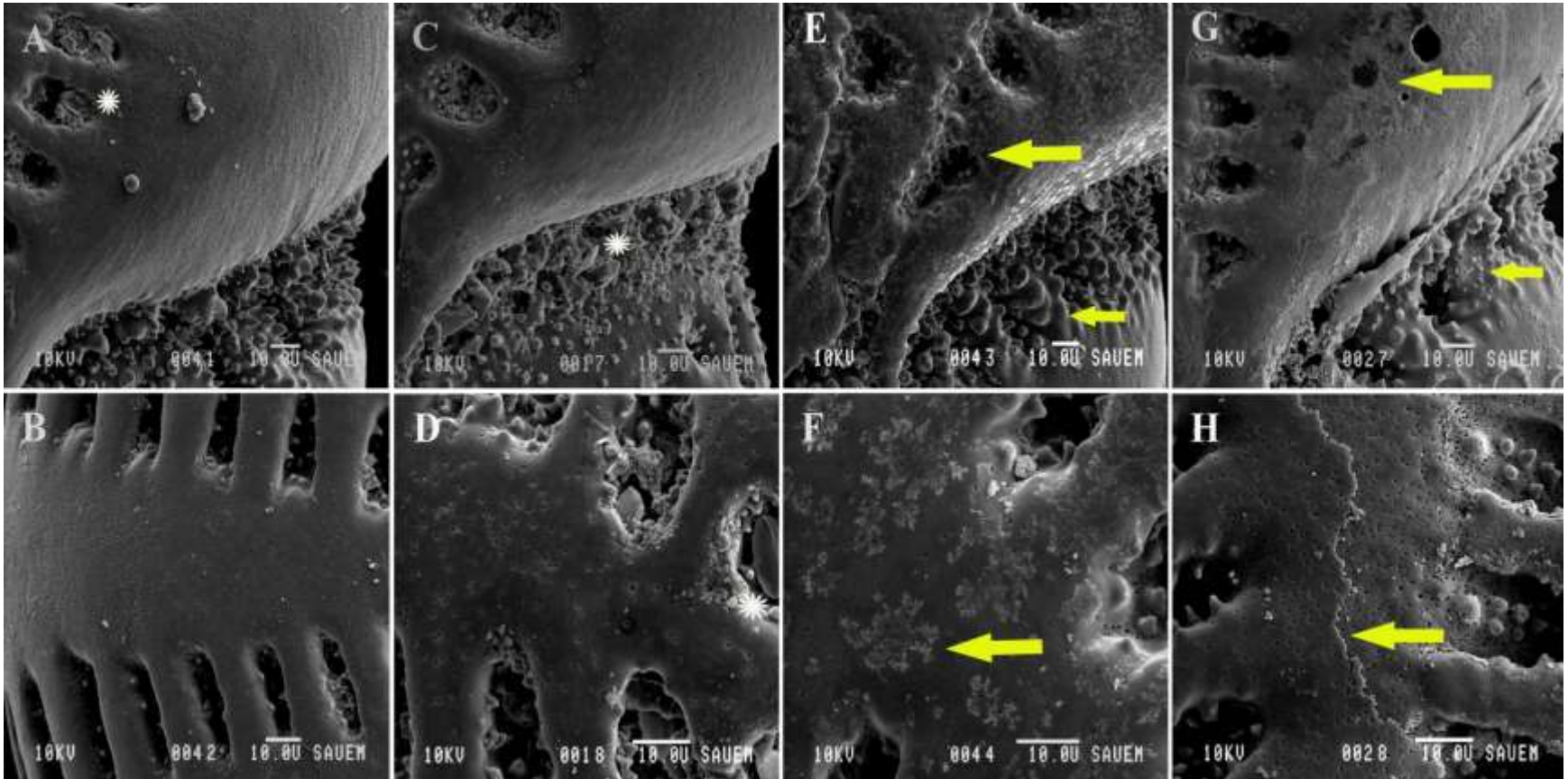
OCEAN ACIDIFICATION

pH 8.1

pH 7.9

pH 7.7

pH 7.3



WHETHER GEOLOGISTS COUPLE OF MILLION YEARS INTO THE FUTURE WOULD WANT TREAT THE ANTHROPOCENE AS THE START OF AN ERA OR AN EPOCH ...

- ...would depend on how fast we get out of the Anthropocene and **how many of the species that existed before the start of the Anthropocene we leave behind.**
- Remember: Each Era starts with a major mass extinction. **If more than 50% of all species end up extinct by the time the 6th mass extinction is over,** the geologists (if there still are any left) of the future may decide to treat the Anthropocene as the beginning of a new Era.
- It is in our best interest to make sure it stays an Epoch in a period called Quaternary which belongs to the Cenozoic era rather than becoming the first Period in a new Era!

WHAT IS THE ANTHROPOCENE AND WHEN DID IT START?

- One single species among many millions of species on mother Earth has recently taken over the planetary driving seat, and now determines its future journey.
- **We, human beings, have become a geological force. Human action is now influencing every aspect of the Earth, at a scale akin to the great forces of nature.**
- There are now so many of us using so much resources, and changing so profoundly the living ecosystems on Earth that we're disrupting the grand cycles of biology, chemistry, and geology.
- Almost all of planet's ecosystems bear the marks of our presence.
- Unsustainable patterns of production, consumption, and population growth are challenging the resilience and the stability of the planet to support human activities.

THE GREAT ACCELERATION

SOCIO-ECONOMIC TRENDS



EARTH SYSTEM TRENDS



REFERENCE: Steffen, W., W. Broadgate, L. Deutsch, O. Gaffney and C. Ludwig, The Trajectory of the Anthropocene: the Great Acceleration, *The Anthropocene Review*, 16 January 2015.

MAP & DESIGN: Félix Pharand-Deschênes / Globaia

The Precautionary Principle: A guiding mantra for addressing Environmental problems

- In 1992, at the Rio Earth summit, it was realized that even with careful scientific research, **it can be difficult or even impossible to prove with absolute certainty how relationships between human activities and other physical and biological processes lead to local and global environmental problems**
- So the Earth summit for Sustainable Development listed as one of its principles what is called “**The Precautionary Principle**”
- **It says that when there is a threat of serious perhaps irreversible environmental damage, we should not wait for scientific proof before taking precautionary steps to prevent potential harm to the environment.**
- This requires critical thinking and the biggest challenge is to decide how much of proof is sufficient to invoke the **Precautionary Principle**

History of Modern Environmental Science

- Before 1960, the term “environment” meant little as a social or political issue
 - In the 1960s several events drew world attention to environmental issues :
 - Oil spills in coastal California,
 - Episodes of photochemical smog and acid rain in Los Angeles
 - Highly publicized threats of extinction to many species : elephants, whales and songbirds
- A book called *Silent Spring* by Rachel Carson

Suddenly environment was a very popular topic with battle lines being drawn between opposing groups

There was very little data to support or negate rumours

Some people even saw Science as part of the problem

WHEN THE EXPERIMENT YOU ARE DOING IS CALLED 6TH MASS EXTINCTION...

- It may be worthwhile to evoke the precautionary principle.
- It easy to be sure that a mass extinction happened by the time 50% of all species on the planet have gone extinct
- However, that threshold will only be reached after the food chains that our very existence depends on, and the marine organisms that our oxygen supply depends on have collapsed...
- ... so we will be one of the species that are unlikely to make it through the extinction event.